

B.Tech in Electronics and Computer Engineering
Semester : 3rd Subject: Data Structures with C (BUE302)
DSC Assignment 1

Ay:25-26

Date of Submission:17-11-2025

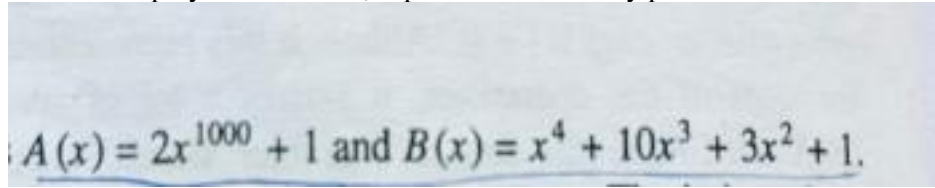
1. Explain the pointer declaration, initialisation and Dereferencing with an example.
2. Explain the following along with their syntax :
(a) malloc(). (b) calloc(). (c) realloc()
3. Explain the allocation and Deallocation of memory with an example program.
4. Define the following
(A) Typecast expression (B) Dangling reference
5. Explain ADT Array
6. Explain the concept of Arrays in C with an example program. ***or*** Explain the implementation of one dimensional arrays with an example.
7. Explain the concept of two dimensional Array with an example.
8. Write a function that creates a two dimensional array at runtime. ***or*** write a function to dynamically create a two dimensional array.
9. Define a structure with the example.
10. What do you mean by typedef explain with an example.
11. Using typedef, write a C program to define a structure **employee** with members as name, age and salary. Then read details of two employees and display the name, salary and age of the employee.
12. Using the concept of embedded structures, write a C program to define a main structure as **Human** with name salary and date of birth as their members. Then display print the DOB of the person.
13. Write the syntax for declaring and defining the union in C and also illustrate the usage of a union with an example.
14. Differentiate between a structure and the union.
15. What is internal implementation of structures?
16. Define self referential structures with an example.
17. Explain ADT polynomial.
18. Explain polynomial representation in C using typedef and also write an alternate representation to preserve space.

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19. Given a polynomial below, Explain with the Array presentation.



$A(x) = 2x^{1000} + 1$ and $B(x) = x^4 + 10x^3 + 3x^2 + 1.$

20. Define Sparse Matrix with an example.
21. Explain ADT Sparse matrix.
22. Explain Sparse Matrix representation.
23. Write a program for fast transpose matrix.
24. Explain stack with a neat diagram .
25. Explain ADT Stack.
26. Explain queue with a neat diagram.
27. Explain ADT Queue.
28. Explain circular queue with a neat diagram.
29. Write a C program to implement circular queues using dynamically allocated using and perform the following operations
i. insert (), ii. Delete() iii. Display()
30. Write a C program to perform iterative and recursive binary search.
31. Write a program for the multiplication of matrix.
32. Define a string and give the memory representation of a string with an example.
33. Explain ADT String
34. Explain string insertion by memory representation.
35. Write a function for string insertion.
36. Write any 5 **string Function** in C.
37. Write a function for **pattern matching by checking end indices first**
38. With a neat figure show, how the function of **pattern matching by checking and indices first** compares, characters from pattern with those of string, Supposing **pat** = "aab" and **string** = "ababbaabaa."
39. Write a program to perform addition of two polynomials.
40. Write a program for transpose Matrix

*****End of Assignment 1*****